CSE4104 Week 01 Project Report

Official Team Information

- **Official Team Name:** CSE4104-7C-T04
- **Section:** 7C
- **Project Title:** AI Code Security Analyzer

Team Information

Team Leader

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Team Members

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Preliminary Project Description

Pluto AI aims to modernize software security analysis by combining advanced artificial intelligence models with automated code review techniques. The system will help developers identify vulnerabilities, insecure coding practices, authentication flaws, and logic weaknesses directly from source code.

The platform will support multiple programming languages and allow users to upload code files or repositories for automated analysis. After processing, the system will generate detailed vulnerability reports, severity analysis, explanations, and remediation suggestions in natural language.

Unlike traditional static analysis tools that rely mostly on predefined signatures and fixed rules, Pluto AI will leverage multiple large language models to understand the semantic structure and logical flow of code. This enables the system to detect deeper vulnerabilities and provide context-aware security recommendations.

The project primarily targets:

- Software developers

- Programming students
- Freelance developers
- Startups and software companies
- Cybersecurity learners

The system will also improve learning by explaining why vulnerabilities occur and how secure coding practices can prevent them.

Project Overview

Pluto AI is a web-based AI-powered code security analyzer that automatically detects vulnerabilities such as SQL injection, XSS, hardcoded secrets, and authentication flaws in source code using multiple large language models.

Key Objectives

- Detect security vulnerabilities automatically
- Provide intelligent remediation suggestions
- Support multiple programming languages
- Generate professional security analysis reports
- Improve developer productivity and secure coding practices

Why AI is Needed in This Project

Traditional code analyzers mainly depend on static rule-based scanning, which often fails to detect complex vulnerabilities, insecure business logic, or context-dependent security flaws. Manual code review by humans is also expensive, time-consuming, and inconsistent.

Artificial intelligence enables Pluto AI to analyze code intelligently by understanding patterns, developer intent, and logical relationships within the source code. Using multiple AI providers allows the system to compare responses and improve analysis reliability.

AI is essential in this project because it can:

- Detect hidden or context-aware vulnerabilities
- Analyze multiple programming languages efficiently
- Generate understandable explanations for developers
- Provide automatic remediation suggestions

- Reduce human effort and review time
- Improve software security awareness
- Assist beginner programmers in learning secure coding practices

By integrating generative AI models, the platform can provide smarter, faster, and more adaptive security analysis compared to conventional systems.

AI Advantages

- Semantic understanding of source code
- Intelligent vulnerability detection
- Natural language explanations for security flaws
- AI-generated remediation suggestions
- Faster and scalable security analysis

Initial Technology Stack

Frontend

- React.js + Vite

Backend

- FastAPI (Python)

Database

- PostgreSQL
- MongoDB
- phpMyAdmin

AI Tools / APIs

- Claude (Anthropic)
- GPT-4o (OpenAI)
- Gemini (Google)
- DeepSeek

Report Formats

- JSON
- Markdown
- PDF

Similar Existing Systems / Research

Several existing tools are currently used for software security analysis and vulnerability scanning. However, most of them mainly rely on predefined rules, signature matching, or static analysis engines. Pluto AI attempts to improve this process through intelligent semantic analysis using multiple AI models.

Existing tools like Snyk, SonarQube, and Semgrep perform static analysis but mostly rely on rule-based engines. Pluto AI differentiates itself by leveraging generative AI models for deeper semantic understanding, multi-provider flexibility, and natural language remediation suggestions.

ExistingSystem	**Features**	**Limitations**	**Possible Improvements**
Snyk	Security vulnerability scanning	Mostly rule-based analysis	AI-powered semantic analysis
SonarQube	Static code quality analysis	Limited contextual understanding	Natural language remediation
Semgrep	Pattern-based vulnerability detection	Requires manual rule management	Multi-LLM intelligent detection